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COS 472
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Assignment 4

In this assignment, the goal was to implement a DNN with multiple configurations in order to achieve the best results when predicting the target set, 'Experience Level' of a gym-goer from the *Gym Members Exercise Tracking* dataset. Key features of the data set are as follows: The target feature, *Experience level* contained three classifications: 1,2,3; therefore this is a multi-class classification problem. I performed one-hot encoding on all categorical features - this resulted in 16 predictor features. After converting categorical data, the data set was split into Training, Testing and Validation tests with 80%, 10% and 10% splits respectively.

Next, the DNN was implemented and tested with six different configurations. The best performing model was 'Model 6' with the following configurations:

Epoch = 250
Batch size = 8
Structure:[16,8,8,8,3]

The general trends that can be observed from the results are as follows: The models with higher epochs, more hidden layers, and smaller batch sizes generally performed much better than the than other models. More nodes per layer did not

produce better results with this data set. For instance, models 7 and 8 both used 64 nodes per layer, but produced significantly worse results across all sets when compared to using 8 nodes per layer. This does not indicate overfitting, as the accuracy of the training sets is too low. Rather, I would assume that the model is under fitted to the data. A hypothesis or question that I have would be in what cases do more nodes per layer produce better results? My prediction is that more data (perhaps thousands or millions of rows) would be required to best utilize more nodes per layer.

Data

Model	Configurations	Training Accuracy	Validating Accuracy	Testing Accuracy
1	epoch: 100 batch size: 16 structure:[16,8,3]	41.77%	43.30%	39.80%
2	epoch: 100 batch size: 32 structure:[16,8,3]	41.77%	43.30%	39.80%
3	epoch: 100 batch size: 16 structure:[16,8,8,3]	77.51%	80.41%	82.65%
4	epoch: 100 batch size: 8 structure:[16,8,8,3]	77.76%	83.51%	82.65%
5	epoch: 250 batch size: 8 structure:[16,8,8,3]	60.15%	63.92%	65.31%
6	epoch: 250 batch size: 8 structure: [16,8,8,8,3]	78.79%	88.66%	84.69%
7	epoch: 250 batch size: 8 structure: [16,64,64,64,3]	41.77%	43.30%	39.80%
8	epoch: 250 batch size: 8 structure:[16,64,3]	41.77%	43.30%	39.80%

Run 1

```
/Users/colby/Desktop/git/fall2024/COS472/Assignment4/py_files/CustomDNN.py:16: UserWarning: Implicit dimension choice for softmax has been deprecated. Change the call to include dim=X as an argument.  
  x = F.softmax(self.fc3(x))  
Epoch [100/100], Training Accuracy: 41.77%  
Epoch [100/100], Validation Accuracy: 43.30%  
Epoch [100/100], Testing Accuracy: 39.80%
```

Run 2

```
/Users/colby/Desktop/git/fall2024/COS472/Assignment4/py_files/CustomDNN.py:16: UserWarning: Implicit dimension choice for softmax has been deprecated. Change the call to include dim=X as an argument.  
  x = F.softmax(self.fc3(x))  
Epoch [100/100], Training Accuracy: 41.77%  
Epoch [100/100], Validation Accuracy: 43.30%  
Epoch [100/100], Testing Accuracy: 39.80%
```

Run 3

```
/Users/colby/Desktop/git/fall2024/COS472/Assignment4/py_files/CustomDNN.py:16: UserWarning: Implicit dimension choice for softmax has been deprecated. Change the call to include dim=X as an argument.  
  x = F.softmax(self.fc3(x))  
Epoch [100/100], Training Accuracy: 77.51%  
Epoch [100/100], Validation Accuracy: 80.41%  
Epoch [100/100], Testing Accuracy: 82.65%
```

Run 4

```
/Users/colby/Desktop/git/fall2024/COS472/Assignment4/py_files/CustomDNN.py:16: UserWarning: Implicit dimension choice for softmax has been deprecated. Change the call to include dim=X as an argument.  
  x = F.softmax(self.fc3(x))  
Epoch [100/100], Training Accuracy: 77.76%  
Epoch [100/100], Validation Accuracy: 83.51%  
Epoch [100/100], Testing Accuracy: 82.65%
```

Run 5

```
/Users/colby/Desktop/git/fall2024/COS472/Assignment4/py_files/CustomDNN.py:16: UserWarning: Implicit dimension choice for softmax has been deprecated. Change the call to include dim=X as an argument.  
  x = F.softmax(self.fc3(x))  
Epoch [250/250], Training Accuracy: 60.15%  
Epoch [250/250], Validation Accuracy: 63.92%  
Epoch [250/250], Testing Accuracy: 65.31%
```

Run 6

```
W = F.conv2d(x, W, padding=1)  
/Users/colby/Desktop/git/fall2024/COS472/Assignment4/py_files/CustomDNN.py:18: UserWarning: Implicit dimension choice for softmax has been deprecated. Change the call to include dim=X as an argument.  
  x = F.softmax(self.fc3(x))  
Epoch [250/250], Training Accuracy: 78.79%  
Epoch [250/250], Validation Accuracy: 88.66%  
Epoch [250/250], Testing Accuracy: 84.69%
```

Run 7

```
/Users/colby/Desktop/git/fall2024/COS472/Assignment4/py_files/CustomDNN.py:18: UserWarning: Implicit dimension choice for softmax has been deprecated. Change the call to include dim=X as an argument.  
  x = F.softmax(self.fc4(x))  
Epoch [250/250], Training Accuracy: 41.77%  
Epoch [250/250], Validation Accuracy: 43.30%  
Epoch [250/250], Testing Accuracy: 39.80%
```

Run 8

```
/Users/colby/Desktop/git/fall2024/COS472/Assignment4/py_files/CustomDNN.py:16: UserWarning: Implicit dimension choice for softmax has been deprecated. Change the call to include dim=X as an argument.  
  x = F.softmax(self.fc2(x))  
Epoch [250/250], Training Accuracy: 41.77%  
Epoch [250/250], Validation Accuracy: 43.30%  
Epoch [250/250], Testing Accuracy: 39.80%
```